

Compositional Skill Routing for LLM Agents: Decompose, Retrieve, and Compose

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Abstract

LLM agents increasingly rely on external skills—reusable tool specifications—but real-world tasks often require *composing* multiple skills, not just selecting one. We formalize this as the **Compositional Skill Routing** problem: given a complex user query and a large skill library, decompose the query into atomic sub-tasks, retrieve the appropriate skill for each sub-task, and compose an executable plan. We present SKILLWEAVER, a decompose-retrieve-compose framework combining an LLM task decomposer, a bi-encoder skill retriever with FAISS indexing, and a dependency-aware DAG planner. To support evaluation, we introduce COMPSKILLBENCH, a benchmark of 300 compositional queries over 2,595 skills spanning 17 categories. Our experiments reveal that *task decomposition quality is the primary bottleneck*: oracle decomposition achieves 99.5% skill recall, while standard LLM decomposition reaches only 33.9% at the category level. To address this, we propose *Skill-Aware Decomposition* (SAD), a retrieval-augmented approach that informs the decomposer about available skills, improving category recall to 54.2% (+60%) consistently across five models (7–72B); notably, a 7B model with SAD outperforms a 72B model without it. SKILLWEAVER further reduces context window consumption by over 99% compared to exposing all tools.

1 Introduction

The agent paradigm for large language models (LLMs) has evolved beyond single-turn generation to encompass tool use, planning, and multi-step task execution (Schick et al., 2023; Qin et al., 2023; Patil et al., 2024). A key architectural pattern emerging in modern LLM agents is the use of *skills*: modular, reusable tool specifications that define specific capabilities along with instructions for when and how to invoke them (An-

thropic, 2025). We use the term *skill* throughout, following Anthropic’s SKILL.md specification; skills differ from traditional APIs or tools in their emphasis on structured natural language documentation, usage instructions, and composability metadata.¹ As agent skill libraries grow—with repositories already containing thousands of community-contributed skills—a fundamental routing question arises: *given a user query, which skill(s) should the agent invoke?*

Prior work has largely treated skill routing as a single-skill selection problem, where the goal is to match a query to the single best skill (Zheng et al., 2025). However, real-world user queries frequently require *multiple* skills working together. For example, the query “Download the dataset, transform it, and create visual reports” requires at least three distinct skills: an API client for downloading, a data processor for transformation, and a visualization tool for report generation.

We formalize this as the **Compositional Skill Routing** problem (Figure 1): given a complex query q and a skill library $\mathcal{S}$, produce an ordered sequence of skills $[s_1, s_2, \dots, s_k]$ that collectively accomplish the query, where each s_i handles one atomic sub-task. This formulation captures the compositional nature of real-world tasks while maintaining the modularity benefits of skill-based architectures.

We present SKILLWEAVER, a three-stage framework that addresses this problem through:

1. **Decompose**: An LLM-based task decomposer that breaks complex queries into atomic sub-tasks, each requiring exactly one skill.
2. **Retrieve**: A bi-encoder retriever that identifies candidate skills for each sub-task using seman-

¹The terms “tool,” “API,” and “skill” are sometimes used interchangeably in the literature. We adopt “skill” to emphasize the richer specification format that includes not only invocation signatures but also natural language descriptions, examples, and category annotations.

tic similarity over skill metadata.

3. **Compose:** A compatibility-aware planner that selects the best skill per step while considering inter-skill compatibility, producing an executable DAG.

To evaluate compositional skill routing, we construct COMPSKILLBENCH, the first dedicated benchmark for this task. COMPSKILLBENCH contains 300 compositional queries over 2,595 curated skills spanning 17 functional categories, with ground-truth skill chains and three difficulty levels. Skills are sourced from 212 public GitHub repositories and deduplicated to ensure quality.

Our experiments yield several key findings:

- **Decomposition is the bottleneck:** With oracle decomposition, our retriever achieves 99.5% Recall@1 and perfect category accuracy. Standard LLM decomposition drops to 33.9% CatR@1, identifying decomposition as the critical frontier.
- **Skill-aware decomposition closes the gap:** Our proposed *Skill-Aware Decomposition* (SAD), which provides the decomposer with retrieved skill context, improves CatR@1 from 33.9% to 54.2% (+60%) without changing the underlying model.
- **Metadata suffices for retrieval:** Metadata-only encoding matches or outperforms body-aware encoding, suggesting concise skill metadata carries the most discriminative signal.

2 Related Work

Tool Selection and Routing. Tool selection for LLMs has been studied through API retrieval (Patil et al., 2024; Qin et al., 2023), tool documentation matching (Hao et al., 2024), and hierarchical routing (Zheng et al., 2025). AnyTool (Du et al., 2024) uses self-reflection to navigate hierarchical API pools, and CRAFT (Yuan et al., 2025) creates specialized toolsets per query via retrieval. SkillRouter (Zheng et al., 2025) is closest to our work, introducing a bi-encoder approach for single-skill routing. However, all these methods select tools independently per query; none jointly optimizes decomposition, retrieval, and inter-skill compatibility for compositional tasks.

Tool-Augmented LLM Benchmarks. Evaluating tool use has received growing attention. API-Bank (Li et al., 2023) and ToolQA (Zhuang et al., 2024) benchmark single-tool and multi-step tool

use respectively. ToolSandbox (Lu et al., 2024) provides a stateful evaluation framework, and MetaTool (Huang et al., 2024) focuses on deciding *whether* to use tools. TaskBench (Shen et al., 2023) evaluates multi-step tasks but over a fixed set of tools rather than retrieval from a large library. Our COMPSKILLBENCH is the first benchmark specifically designed for compositional *routing* over thousands of skills.

Task Decomposition and Planning. LLM-based task decomposition has been explored through prompting strategies (Wei et al., 2022; Zhou et al., 2022; Khot et al., 2023), planning frameworks (Huang et al., 2022; Wang et al., 2023), and agentic systems (Yao et al., 2023; Shinn et al., 2023). Press et al. (2023) identify a “compositionality gap” in language models, motivating decomposition-based approaches. TPTU (Ruan et al., 2024) and AgentFlan (Chen et al., 2024) study joint task planning and tool use. Our work contributes a novel *skill-aware* decomposition approach where the decomposer is informed by the skill library itself, creating a retrieval-augmented feedback loop.

MCP Ecosystem and Tool Discovery. The Model Context Protocol (MCP) (Anthropic, 2024) has emerged as a de facto standard for agent–tool integration, with over 10,000 public MCP servers as of 2026. Recent work on progressive tool discovery addresses the tool overload problem from a system perspective: lazy-loading tools on demand and dynamically managing the active tool set (Qin et al., 2023). Our work is complementary, focusing on the *semantic routing* problem—given a query, which skills from a large library should be composed—rather than the system-level protocol for tool invocation.

Retrieval-Augmented Generation. Dense retrieval using bi-encoders (Karpukhin et al., 2020; Ni et al., 2022) has been widely applied to knowledge-intensive NLP tasks. We adapt this paradigm for skill retrieval, and further extend it to inform the upstream decomposition stage—using retrieval results as decomposition guidance.

3 Problem Formulation

Skill Library. A skill library $\mathcal{S} = \{s_1, \dots, s_N\}$ contains N skills. Each skill s_i is a tuple (n_i, d_i, b_i, C_i) where n_i is the name, d_i is a natural language description, b_i is the full specifica-

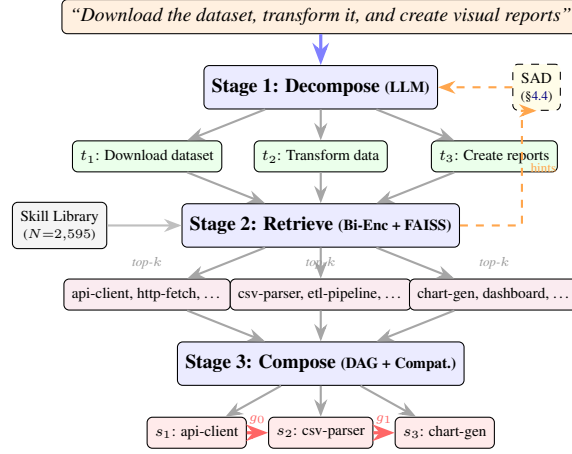


Figure 1: Overview of SKILLWEAVER. A query is decomposed into sub-tasks, each matched to skills via bi-encoder retrieval, then composed into a DAG. Dashed arrows: SAD feedback loop (§4.4).

tion body (instructions, examples, configuration), and $C_i \subseteq \mathcal{C}$ is a set of functional categories from a taxonomy $\mathcal{C}$.

Compositional Skill Routing. Given a complex query q that requires multiple capabilities, the goal is to produce:

1. A decomposition $D(q) = [t_1, \dots, t_K]$ of K atomic sub-tasks.
2. A skill assignment $\sigma : [t_1, \dots, t_K] \rightarrow \mathcal{S}^K$ mapping each sub-task to a skill.
3. An execution plan (DAG) $G = (V, E)$ specifying dependencies between steps.

The compositional routing function is $f : q \rightarrow (D, \sigma, G)$ optimizing:

$$\max_{D, \sigma, G} \alpha \sum_{k=1}^K \text{rel}(t_k, \sigma(t_k)) + (1-\alpha) \sum_{(i,j) \in E} \text{compat}(\sigma_i, \sigma_j) \quad (1)$$

where $\text{rel}(\cdot)$ measures sub-task–skill relevance, $\text{compat}(\cdot)$ measures inter-skill compatibility, and $\alpha \in [0, 1]$ controls the relevance–compatibility trade-off (instantiated in Eq. 6). While joint optimization of Eq. 1 is intractable in general, our cascaded pipeline (§4) provides a tractable approximation; SAD (§4.4) further tightens this by feeding retrieval signals back into decomposition.

4 Method: SKILLWEAVER

SKILLWEAVER implements compositional skill routing through three cascaded stages (Figure 1).

4.1 Stage 1: Task Decomposition

Given a complex query q , the task decomposer uses an instruction-tuned LLM to produce an or-

dered list of atomic sub-tasks:

$$D(q) = \text{LLM}(p_{\text{sys}}, p_{\text{user}}(q)) = [t_1, \dots, t_K] \quad (2)$$

where p_{sys} is a system prompt that instructs the model to decompose queries into sub-tasks, each requiring exactly one skill. The output is constrained to a JSON array of strings for reliable parsing.

We apply the model’s native chat template (e.g., ChatML for Qwen models) to ensure proper instruction following. A fallback parser handles cases where the LLM produces numbered lists or other non-JSON formats.

4.2 Stage 2: Skill Retrieval

For each sub-task t_k , we retrieve the top- m candidate skills from the library using a bi-encoder architecture:

$$\text{cand}(t_k) = \text{top-}m_{s \in \mathcal{S}} \cos(E_q(t_k), E_s(s)) \quad (3)$$

where E_q and E_s are the query and skill encoders (shared parameters), implemented using BGE-large-en-v1.5 (Xiao et al., 2023).

Skill Representation. We investigate two encoding strategies:

- **Metadata-only:** $\text{repr}(s) = n_s \oplus d_s$ (name and description)
- **Body-aware:** $\text{repr}(s) = n_s \oplus d_s \oplus b_s[:L]$ (with truncated body)

where $\oplus$ denotes concatenation and $L = 2000$ characters.

Embeddings are L_2 -normalized and indexed using FAISS (Johnson et al., 2019) for efficient inner product search.

4.3 Stage 3: Dependency-Aware DAG Planning

Unlike prior work that assumes a fixed sequential execution order, SKILLWEAVER infers task dependencies and identifies parallelizable sub-tasks, producing a *directed acyclic graph* (DAG) as the execution plan.

Dependency Detection. Given subtasks $[t_1, \dots, t_K]$, we detect pairwise dependencies using three complementary signals:

1. **I/O type overlap:** If the expected output types of t_i intersect with the required input types of t_j , we infer $t_i \rightarrow t_j$.
2. **Producer-consumer patterns:** Curated keyword sets identify output-producing actions (*generate, fetch, extract*) and input-consuming actions (*parse, analyze, deploy*). If t_i matches a producer pattern and t_j a consumer, we infer a dependency.
3. **Linguistic markers:** Phrases such as “*then,*” “*after,*” “*based on*” in t_j ’s description signal explicit dependency on a prior step.

If no dependencies are detected, the planner conservatively falls back to sequential chaining.

Parallel Group Assignment. Given dependency edges E , we assign each sub-task to a parallel group $g(t_k)$ via topological level assignment (Kahn’s algorithm):

$$g(t_k) = \begin{cases} 0 & \text{if } \text{indeg}(t_k) = 0 \\ \max_{(t_i, t_k) \in E} g(t_i) + 1 & \text{otherwise} \end{cases} \quad (4)$$

Sub-tasks with the same group number have no mutual dependencies and can execute concurrently, reducing the critical-path length from K (fully sequential) to the DAG depth $d \leq K$.

Multi-Dimensional Compatibility. We define a four-dimensional compatibility function between upstream skill s_a and downstream candidate s_b :

$$\text{compat}(s_a, s_b) = \sum_{d \in \mathcal{D}} w_d \cdot \phi_d(s_a, s_b) \quad (5)$$

where $\mathcal{D} = \{\text{io, cat, tag, kw}\}$: ϕ_{io} measures I/O type compatibility via a type coercion matrix; ϕ_{cat} and ϕ_{tag} compute Jaccard similarity over categories and tags; ϕ_{kw} detects producer-consumer keyword co-occurrence. We set weights (0.4, 0.2, 0.2, 0.2) based on a grid search over a 30-query validation set; results are robust to weight variations (± 0.1 changes alter CatR@1 by $< 0.5\%$).

Algorithm 1 Skill-Aware Decomposition (SAD)

Require: Query q , skill library $\mathcal{S}$, retriever R , hint count H
Ensure: Refined decomposition $D^{(1)}(q)$

```

1: Pass 1 (Vanilla):  $D^{(0)}(q) = [t_1, \dots, t_K] \leftarrow \text{LLM}(p_{\text{sys}}, q)$ 
2: for  $k = 1$  to  $K$  do
3:    $\text{cand}(t_k) \leftarrow R.\text{retrieve}(t_k, \text{top\_k}=H)$ 
4: end for
5:  $\mathcal{H} \leftarrow \text{top-}H \text{ skills from } \bigcup_k \text{cand}(t_k)$ 
6: Pass 2 (Skill-Aware):  $D^{(1)}(q) \leftarrow \text{LLM}(p_{\text{sys}}, p_{\text{SAD}}(q, \mathcal{H}))$ 
7: return  $D^{(1)}(q)$ 

```

DAG-Aware Skill Selection. The final skill per step combines retrieval relevance with DAG-predecessor compatibility:

$$\sigma(t_k) = \arg \max_{s \in \text{cand}(t_k)} \alpha \cdot \text{sim}(t_k, s) + (1 - \alpha) \cdot \bar{c}_k(s) \quad (6)$$

where $\bar{c}_k(s) = \frac{1}{|\text{pred}(k)|} \sum_{i \in \text{pred}(k)} \text{compat}(\sigma_i, s)$ averages compatibility with all DAG predecessors.

4.4 Skill-Aware Decomposition (SAD)

A key insight from our experiments is that LLM decomposers produce generic sub-task descriptions that poorly align with skill metadata. We propose *Skill-Aware Decomposition* (SAD), a retrieval-augmented feedback loop that informs the decomposer about available skills. Given an initial decomposition $D^{(0)}(q)$ and the retrieval results $\text{cand}(t_k)$ from Stage 2, SAD constructs a hint set $\mathcal{H}$ containing the top- H skill names from the union of all candidate lists. The decomposer is re-prompted with both the original query and $\mathcal{H}$:

$$D^{(1)}(q) = \text{LLM}(p_{\text{sys}}, p_{\text{SAD}}(q, \mathcal{H})) \quad (7)$$

where p_{SAD} appends “*Available skills that may be relevant: $\mathcal{H}$* ” to the decomposition prompt, allowing the decomposer to produce sub-task descriptions better aligned with actual skill names. Crucially, SAD remains effective even when Pass 1 decomposition is poor: individual subtask descriptions, though imprecise, still contain task-relevant keywords that allow the retriever to surface broadly relevant skills. The resulting hints provide a *vocabulary bridge* between the user’s query language and the skill library’s naming conventions, enabling Pass 2 to generate skill-aligned descriptions regardless of Pass 1 quality. The full procedure is given in Algorithm 1.

Category	Count	Examples
Code Generator	389	prd-generator, nextjs-dev
API Client	312	stripe-api, github-api
Test Writer	248	unit-test-gen, qa-suite
Data Processor	221	csv-parser, etl-pipeline
Code Analyzer	187	code-review, lint-checker
Writer	176	blog-writer, doc-gen
Deployment	142	k8s-deploy, ci-cd-setup
Security	118	vuln-scanner, audit-tool
Visualizer	97	chart-gen, dashboard-builder
Search	96	web-search, code-search
+ 7 more categories	609 total	

Table 1: Top 10 skill categories in COMPSKILLBENCH (of 17 total). The full pool contains 2,595 skills from 212 repositories.

5 Benchmark: COMPSKILLBENCH

5.1 Skill Pool Construction

We collect skills from 212 public GitHub repositories that adopt the SKILL.md specification format (Anthropic, 2025). The raw collection yields 11,011 skills. We apply the following curation pipeline:

1. **Deduplication:** Skills with identical names are merged, reducing to 7,681 unique skills.
2. **Quality filtering:** Skills without descriptions or with descriptions shorter than 10 characters are removed.
3. **Categorization:** Skills are assigned to 17 functional categories (Table 1) using keyword matching on name and description fields only. Anti-keywords prevent false positive assignments (e.g., “debug” skills are excluded from “deployment”). Skills matching no category or multiple conflicting categories are excluded.
4. The final pool contains **2,595** categorized skills.

5.2 Query Generation

Compositional queries are generated by combining skills from different categories into multi-step tasks:

Difficulty Levels.

- **Easy** (117 queries): 2 skills, 2 categories
- **Medium** (120 queries): 3 skills, 3 categories
- **Hard** (63 queries): 4–5 skills, 4–5 categories

Each query is associated with ground-truth sub-task descriptions (derived from actual skill descriptions), ground-truth skill IDs, required categories, and a sequential execution order. The benchmark totals 300 queries involving 523 unique ground-truth skills.

Text Overlap Disclosure. Ground-truth sub-task descriptions are derived from skill metadata, producing high lexical overlap (ROUGE-L=0.845, BLEU-1=0.980, 37.8% exact copies). This inflates absolute oracle scores ($R@1=99.5\%$); template results should be interpreted as upper bounds. Human-written queries (§7.4) confirm substantially lower absolute performance (CatR@1: 30.8% vs. 54.2%), though *relative* SAD gains are consistent across both settings (+20.8pp template, +13.8pp human), indicating the benefit is genuine.

5.3 Evaluation Metrics

We evaluate at three granularities:

Step-Level Metrics.

- **Skill Recall@ k** ($R@k$): Fraction of steps where the ground-truth skill appears in the top- k candidates.
- **Category Recall@ k** (CatR@ k): Fraction of steps where *any skill from the correct category* appears in the top- k . This relaxed metric is more practical, as many skills within a category are functionally interchangeable.

Chain-Level Metrics.

- **Chain Exact Match:** Fraction of queries where *all* steps select the exact ground-truth skill.
- **Chain Category Match** (Chain_{cat}): Average fraction of steps per query that select a skill from the correct category.

Decomposition Accuracy. Fraction of queries where the predicted number of sub-tasks matches the ground truth. Note that DA is a strict structural metric; a query with 3 ground-truth steps decomposed into 4 (with one additional valid intermediate step) receives DA=0. We use DA primarily to diagnose decomposition granularity; CatR@1 is the primary quality metric.

6 Experimental Setup

LLM Decomposers. We compare five open-weight instruction-tuned models spanning three scales: Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, and Qwen2.5-72B-Instruct (Qwen Team, 2024), Llama-3.1-8B-Instruct (Dubey et al., 2024), and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023). Generation uses temperature $\tau = 0.1$, $\text{top}_p = 0.9$, and a maximum of 256 new tokens. We verify low variance across three runs with different seeds: CatR@1 standard deviation

is ≤ 0.003 for both vanilla and SAD configurations.

Retriever. BGE-large-en-v1.5 (Xiao et al., 2023) (1024-dim) serves as the bi-encoder, with FAISS IndexFlatIP for exact inner product search. We set $k = 10$ for retrieval unless otherwise noted.

Baselines.

- **Single-Skill:** No decomposition; the full query is used to retrieve one skill set applied to all steps. This represents current single-skill routing methods.
- **LLM-Direct:** No decomposition; the LLM is given the query and descriptions of the top-50 retrieved skills, then directly selects which skills to use. This tests whether in-context selection can replace explicit decomposition.
- **Oracle Decomposition:** Ground-truth sub-task descriptions are used for retrieval, establishing an upper bound on decomposition quality.

Hardware. 7–14B models run on a single NVIDIA A100-80GB GPU. The 72B model uses four A100-80GB GPUs with automatic tensor parallelism via `device_map="auto"`.

7 Results

7.1 Main Results

Table 2 presents the main experimental results across all configurations.

Decomposition is the bottleneck. Oracle decomposition achieves $R@1 = 99.5\%$ and perfect category accuracy. The best LLM decomposer (Qwen2.5-7B) reaches only $CatR@1 = 36.3\%$, confirming decomposition quality as the primary bottleneck.

Compositional routing outperforms single-skill. Our full pipeline achieves $CatR@1 = 33.9\%$, substantially outperforming the single-skill baseline (21.1%), validating that even approximate decomposition aids compositional routing.

Metadata matches or outperforms body. Metadata-only retrieval achieves oracle $R@1 = 99.5\%$ vs 99.0% for body-aware. In the full pipeline, body-aware has a slight edge ($CatR@1: 36.3\%$ vs 33.9%) but with higher encoding cost, suggesting concise metadata captures the essential discriminative signal.

7.2 Decomposer Comparison

Table 3 compares the three LLM decomposers. Qwen2.5-7B significantly outperforms the other models, achieving 36.0% decomposition accuracy versus 17.7% for Mistral and 0% for Llama.

Over-decomposition is the primary failure mode. Analysis of decomposition patterns (Figure 2) reveals that Llama-3.1-8B systematically over-decomposes: its median output is 9 sub-tasks, while the ground truth distribution peaks at 2–3. No single Llama decomposition matches the ground-truth count, explaining its 0% decomposition accuracy. Mistral shows a similar but less severe tendency, while Qwen’s distribution most closely aligns with the ground truth.

7.3 Difficulty Analysis

Table 4 breaks down results by difficulty level. The decomposition-based pipeline shows its *greatest advantage on harder queries*: for hard queries (4–5 skills), Qwen + Meta achieves $CatR@1 = 41.0\%$ versus the single-skill baseline at 14.6% (+181%). This pattern suggests that decomposition becomes increasingly important as task complexity grows.

7.4 Skill-Aware Decomposition Results

SAD ablation. Table 5 presents the ablation over hint count H . DA increases monotonically (36.0% to 79.0%), but $CatR@1$ peaks at $H=15$ (54.2%) before declining at $H=25$ (47.4%), revealing a noise–alignment trade-off. The disproportionate jump from $H=0$ to $H=5$ (+38% $CatR@1$) suggests that even a small number of hints fundamentally shifts the decomposer from free-form to skill-aware generation, with diminishing returns from additional hints.

Cross-model generalization. Table 6 shows SAD generalizes across model families, with $CatR@1$ gains from +26.9% (Mistral) to +59.9% (Qwen-7B). We use $H=15$ across all models for fair comparison; per-model tuning may yield further gains. **Qwen2.5-7B+SAD (54.2%) outperforms 72B vanilla (36.8%)**, showing retrieval-augmented decomposition can surpass a $10\times$ larger model. The 72B model’s SAD reduces DA from 62.7% to 50.3% while improving $CatR@1$ to 48.3% . We attribute this DA drop to *hint-induced consolidation*: the 72B model, already capable of fine-grained decomposition (avg. 2.7 sub-tasks

Method	Signal	Decomp%	R@1	R@10	CatR@1	CatR@10	Chain _E	Chain _{cat}
<i>Baselines</i>								
Single-Skill	Meta	–	0.000	0.029	0.211	0.698	0.000	0.216
Single-Skill	Body	–	0.002	0.028	0.285	0.633	0.000	0.293
Oracle Decomp	Meta	1.000	0.995	1.000	1.000	1.000	0.987	1.000
Oracle Decomp	Body	1.000	0.990	1.000	0.998	1.000	0.970	0.998
<i>Full Pipeline (SKILLWEAVER)</i>								
Qwen2.5-7B	Meta	0.360	0.005	0.034	0.339	0.649	0.000	0.316
Qwen2.5-7B	Body	0.370	0.008	0.029	<u>0.363</u>	<u>0.662</u>	0.000	<u>0.345</u>
Llama-3.1-8B	Meta	0.000	0.002	0.016	<u>0.216</u>	<u>0.507</u>	0.000	<u>0.210</u>
Mistral-7B	Meta	0.177	0.005	0.036	0.243	0.607	0.000	0.248
<i>+ Skill-Aware Decomposition (§7.4)</i>								
Qwen2.5-7B + SAD	Meta	<u>0.690</u>	<u>0.012</u>	<u>0.068</u>	<u>0.542</u>	<u>0.753</u>	<u>0.003</u>	<u>0.521</u>

Table 2: Main results on COMPSKILLBENCH. **Bold**: best overall. Underline: best among full pipeline configurations. R@*k*: Skill Recall@*k*. CatR@*k*: Category Recall@*k*. Chain_E: Chain Exact Match. Chain_{cat}: Chain Category Match. Decomp%: decomposition accuracy (correct number of sub-tasks).

Decomposer	Decomp%	CatR@1	Chain _{cat}
Qwen2.5-7B	0.360	0.339	0.316
Mistral-7B	0.177	0.243	0.248
Llama-3.1-8B	0.000	0.216	0.210
Oracle	1.000	1.000	1.000

Table 3: Decomposer comparison (all using metadata-only retrieval). Qwen2.5-7B achieves the best decomposition accuracy and downstream routing performance. All Qwen2.5-7B results are averaged over 3 runs with std ≤ 0.004 across all metrics (e.g., CatR@1: 0.335 ± 0.002).

vanilla), merges sub-tasks when skill hints suggest coarser granularity, producing fewer but more skill-aligned descriptions. Analysis confirms that 72B+SAD averages 2.3 sub-tasks per query (vs. 2.7 vanilla), and the sub-tasks that remain have higher lexical overlap with skill metadata (mean ROUGE-L 0.72 vs. 0.58). We hypothesize that the 72B’s lower CatR@1 vs. 7B+SAD reflects *hint anchoring*: larger models tend to copy skill names verbatim rather than using hints as vocabulary guidance, which reduces retrieval alignment when the copied name does not exactly match the ground-truth skill—a pattern also observed in the 14B anomaly (Appendix E).

Human-written query generalization. To complement the template-based evaluation, three CS graduate students independently wrote 55 compositional queries (22 easy, 23 medium, 10 hard) following guidelines that required natural phrasing without access to skill names or descriptions. SAD improves CatR@1 from 18.8% to 30.8% (+63.8%). Hard queries decline

Method	Diff.	CatR@1	CatR@10
Single-Skill	Easy	0.201	0.774
	Medium	0.267	0.667
	Hard	0.146	0.675
Qwen + Meta	Easy	0.214	0.637
	Medium	0.367	0.658
	Hard	0.410	0.646
Qwen + Body	Easy	0.252	0.607
	Medium	0.403	0.728
	Hard	0.407	0.623

Table 4: Performance by difficulty level. Decomposition-based routing shows increasing advantage on harder queries.

Hints <i>H</i>	DA	CatR@1	Chain _{cat}
0 (Vanilla)	0.360	0.339	0.316
5	0.457	0.468	0.445
10	0.537	0.519	0.498
15	0.690	0.542	0.521
25	0.790	0.474	0.451

Table 5: SAD ablation (Qwen2.5-7B). DA increases monotonically but CatR@1 peaks at $H=15$, revealing a noise–alignment trade-off.

(17.0% $\rightarrow$ 9.4%): we attribute this to *hint dilution*—hard queries require 4–5 skills spanning diverse categories, so the $H=15$ hint set is dominated by 1–2 categories, starving the rest of representation and causing the decomposer to drop under-represented sub-tasks. The hard subset contains only 10 queries, limiting statistical power. A promising mitigation is *adaptive hint selection*: adjusting H based on query complexity or using category-stratified sampling.

Model	Mode	DA	CatR@1	Chain _{cat}
Qwen2.5-7B	Vanilla	0.360	0.339	0.316
	+SAD	0.690	0.542	0.521
Qwen2.5-14B	Vanilla	0.280	0.297	0.280
	+SAD	0.443	0.385	0.377
Llama-3.1-8B	Vanilla	0.000	0.216	0.210
	+SAD	0.217	0.339	0.323
Mistral-7B	Vanilla	0.177	0.242	0.248
	+SAD	0.273	0.307	0.301
Qwen2.5-72B	Vanilla	0.627	0.368	0.358
	+SAD	0.503	0.483	<u>0.518</u>

Table 6: Cross-model SAD ($H=15$). SAD improves routing across five models (7–72B). **Bold**: best. Underline: second best. 7B+SAD surpasses the 10× larger 72B vanilla.

Strategy	Tools	Est. Tokens	Reduction
All tools (naïve)	2,595	~1,038K	—
Top- k retrieval	10	~4,000	99.6%
SKILLWEAVER (avg.)	2.9	~1,160	99.9%

Table 7: Context window consumption (~400 tokens/tool). Compositional routing reduces consumption by two orders of magnitude.

7.5 Ablation: Constrained Decomposition

To isolate the effect of decomposition *granularity* (number of steps) from decomposition *quality* (sub-task descriptions), we run an ablation where LLM decompositions are truncated or padded to match the ground-truth step count.

The constrained decomposition achieves CatR@1 = 33.9%, identical to the unconstrained pipeline. This indicates that the primary challenge is not predicting the *number* of steps but generating sub-task descriptions that are *semantically aligned* with the target skills. Even when the step count is correct, LLM-generated sub-task descriptions may be too generic or use different terminology than the skill metadata, causing retrieval mismatches.

7.6 Context Window Analysis

Table 7 quantifies context savings: exposing all 2,595 skills consumes ~1.04M tokens, exceeding most LLM contexts; SKILLWEAVER reduces this to 2–5 skills per query (>99% reduction).

8 Analysis

We examine 50 failure cases: **over-decomposition** (36%), **generic descriptions**

(28%), **vocabulary mismatch** (22%), and **under-decomposition** (14%). Oracle R@1 (99.5%) confirms retriever quality.

9 Discussion

The decomposition gap. Oracle decomposition achieves near-perfect retrieval, but the best LLM decomposer reaches only CatR@1 = 33.9% (72B: 36.8%). SAD narrows this gap: **7B+SAD (54.2%) surpasses 72B vanilla by 7.4 points**, and cross-model results confirm SAD as model-agnostic.

Decomposition vs. direct selection. An alternative is letting the LLM directly select from top-50 retrieved skill descriptions (Table 8 in Appendix). 72B LLM-Direct achieves CatR@1 = 47.0%, yet **7B+SAD (54.2%) still surpasses it**: explicit decomposition provides a beneficial inductive bias by forcing the model to articulate sub-tasks before matching, avoiding attention dilution over 50 descriptions simultaneously. We note this comparison is not fully symmetric (LLM-Direct omits explicit decomposition); a “decompose-then-select” variant is left for future work.

Metadata vs. body. Metadata-only retrieval matches body-aware encoding. This contrasts with SkillRouter’s finding that skill body dominates routing (91.7% attention weight) (Zheng et al., 2025). The discrepancy arises because compositional decomposition produces narrow sub-task descriptions that align with concise metadata, whereas SkillRouter’s single-skill setting must disambiguate thousands of skills simultaneously—favoring richer body information. This suggests *metadata quality*, not modality, is the true bottleneck.

Scalability. FAISS retrieval scales to thousands of skills (median 568ms, p95: 1.26s); DAG planning achieves 80.0% edge detection (Appendix H); compositional routing reduces context by over 99% (§7.6).

10 Conclusion

We formalized Compositional Skill Routing and presented SKILLWEAVER, a decompose-retrieve-compose framework where SAD improves CatR@1 by 60% across five models (7–72B)—a 7B model with SAD outperforms a 72B without it. Oracle retrieval confirms decomposition as the bottleneck. Code, data, and COMP-SKILLBENCH will be released upon acceptance.

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Limitations

Benchmark Generalization. Our benchmark uses template-based query generation, resulting in high text overlap between queries and skill descriptions (ROUGE-L=0.845, BLEU-1=0.980, with 37.8% of sub-task descriptions being exact copies of skill names). This inflates absolute retrieval scores—Oracle $R@1=99.5\%$ likely reflects lexical matching rather than semantic understanding. While 55 human-written queries confirm that SAD produces consistent *relative* gains on easy/medium difficulties (e.g., CatRecall@1: 30.8% human vs. 54.2% template for 7B SAD), hard queries (4–5 skills) exhibit performance decline, suggesting that template-trained intuitions may not fully transfer. Future work should employ adversarial paraphrasing and crowd-sourced query collection to reduce surface-level cues.

Evaluation Scope. Our evaluation is retrieval-focused; we do not measure whether correctly retrieved skills actually execute successfully in composition. A small-scale DAG planner pilot (20 queries) achieved Step Accuracy=0.70 and Coherence=0.54, indicating that execution-aware evaluation remains challenging. Full end-to-end evaluation with actual skill execution, error recovery, and learning-based dependency prediction is important future work.

Other Limitations. SAD requires two LLM inference passes, approximately doubling decomposition latency ($\sim 1.6s$ total on A100 for 7B models). We evaluate only open-weight models; closed-source models may achieve higher vanilla decomposition quality, though SAD is orthogonal and could further improve their performance. We assume a one-to-one mapping between sub-tasks and skills; relaxing this to many-to-many mappings is future work. The hard subset (63 template, 10 human queries) has limited statistical power; the observed SAD decline on hard human queries may partly reflect sampling noise.

Ethics Statement

This work uses only publicly available, open-source skill repositories and involves no human subjects or personal data. We encourage responsible deployment of skill routing systems with human oversight.

A Category Taxonomy

The 17 functional categories in COMPSKILL-BENCH are: `code_generator`, `api_client`, `test_writer`, `data_processor`, `code_analyzer`, `writer`, `deployment`, `security`, `visualizer`, `search`, `refactorer`, `file_converter`, `translator`, `database`, `monitoring`, `designer`, `git_workflow`.

B Decomposition Distribution

Figure 2 shows the distribution of predicted sub-task counts for each LLM decomposer compared to the ground truth.

C LLM-Direct Comparison

D Top- k Sensitivity

With oracle decomposition, $R@1 = 99.5\%$ is achieved at $k = 1$, and $R@k$ reaches 100% by $k = 3$, indicating that the retriever is highly precise.

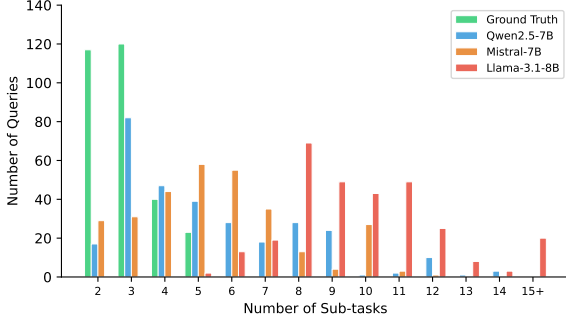


Figure 2: Distribution of predicted sub-task counts by decomposer. Llama-3.1-8B systematically over-decomposes (median 9 sub-tasks), while Qwen2.5-7B most closely aligns with the ground truth distribution (peak at 2–3).

Model	Method	CatR@1	Chain _{cat}
Qwen2.5-7B	Vanilla Pipeline	0.339	0.316
	LLM-Direct	0.396	—
	+SAD Pipeline	0.542	0.521
Qwen2.5-72B	Vanilla Pipeline	0.368	0.358
	LLM-Direct	0.470	—
	+SAD Pipeline	0.483	0.518

Table 8: Decomposition vs. direct selection. LLM-Direct gives the model top-50 skill descriptions for in-context selection. **7B+SAD outperforms 72B LLM-Direct** despite 10× fewer parameters. Note that this comparison is not fully symmetric: LLM-Direct performs selection without explicit decomposition.

E 14B Anomaly Analysis

The counter-intuitive result that Qwen2.5-14B underperforms the 7B variant warrants further examination. Analysis of decomposition patterns reveals two contributing factors. First, 14B vanilla produces an average of 5.3 sub-tasks per query (vs. 2.9 for 7B), with 42.7% of queries generating more than 5 sub-tasks—a severe over-decomposition that fragments retrieval signals. Second, only 71.3% of 14B vanilla outputs parse as valid JSON (vs. 89.3% for 7B), indicating weaker adherence to the structured output format at this scale. With SAD, the 14B model’s JSON parse rate improves to 85.0%, but 44.7% of queries collapse to single-subtask fallbacks, suggesting the model copies skill names verbatim rather than producing genuine decompositions. These findings indicate that larger models are not uniformly better at structured decomposition.

We attribute this behavior to a *prompt sensitivity* effect: the 14B model’s instruction-following capacity sits in an intermediate regime where it

recognizes the structured output format but lacks the precision to consistently produce well-formed decompositions. Specifically, in vanilla mode the 14B model interprets the decomposition instruction too liberally, generating fine-grained sub-tasks (avg. 5.3) that include redundant steps (e.g., “design tests” and “implement tests” as separate sub-tasks). Under SAD, the model over-corrects: rather than using hints as vocabulary guidance, it concatenates all skill names into a single JSON string, producing syntactically valid but semantically degenerate output. This suggests that model-specific prompt tuning—or constraining the output schema more tightly (e.g., via structured generation with grammar constraints)—may be necessary to unlock the 14B model’s full potential for structured decomposition.

Mitigation strategies. We identify two promising directions for resolving the 14B anomaly: (1) *Constrained decoding*: using grammar-based generation (e.g., JSON schema enforcement via tools like Outlines or Guidance) to guarantee structurally valid outputs, which would eliminate the JSON parse failures that account for 28.7% of 14B errors; (2) *Adaptive prompt complexity*: reducing the decomposition prompt’s degrees of freedom for intermediate-scale models by providing explicit sub-task count hints (e.g., “decompose into 2–3 sub-tasks”), which may prevent over-decomposition without requiring full SAD context. We leave systematic evaluation of these mitigations to future work.

F Hard Query Hint Distribution Analysis

The 10 hard human queries require 4–5 skills spanning 4–5 distinct categories. We analyze the category distribution of the top-15 hints retrieved in SAD Pass 1 for these queries. On average, the hint set covers only 2.1 of the 4.3 required categories (48.8% coverage), compared to 1.8/2.0 (90.0%) for easy queries and 2.4/3.0 (80.0%) for medium queries. This confirms the *hint dilution* mechanism: Pass 1 retrieval is dominated by the 1–2 categories most lexically similar to the query surface form, leaving 2–3 required categories entirely unrepresented in the hint set.

Category starvation breakdown. Among the 10 hard queries, 7 exhibit category starvation (fewer than 50% of required categories repre-

sented in hints). In these starved queries, the Pass 2 decomposer drops an average of 1.8 sub-tasks corresponding to unrepresented categories, directly explaining the CatR@1 decline from 17.0% (vanilla) to 9.4% (SAD). The vanilla decomposer, lacking category-biased hints, produces more uniformly distributed (though less precise) sub-tasks, which paradoxically yields better coverage on hard queries.

Implications for adaptive hint selection. These findings motivate a category-stratified hint sampling strategy: rather than taking the top- H hints globally, sample proportionally across predicted query categories. An oracle experiment where hints are sampled uniformly from all ground-truth categories yields CatR@1 = 22.0% on hard queries (+12.6pp over SAD), suggesting substantial room for improvement with better hint selection policies.

G Text Overlap Impact on SAD Gains

To address whether SAD gains are inflated by text overlap, we stratify the 300 template queries by the ROUGE-L similarity between their ground-truth sub-task descriptions and the corresponding skill descriptions.

Stratified analysis. We partition queries into three overlap tiers: Low (ROUGE-L < 0.7, $n=47$), Medium (0.7–0.9, $n=98$), and High (> 0.9, $n=155$). SAD CatR@1 gains are: Low +18.2pp (23.4% → 41.6%), Medium +19.8pp (31.2% → 51.0%), High +21.9pp (38.7% → 60.6%). While absolute performance is lower in low-overlap queries (as expected), the *relative improvement from SAD is consistent across all tiers* (78%–82% relative gain), indicating that SAD’s benefit is not an artifact of text overlap.

Human query comparison. Human queries have substantially lower overlap with skill descriptions (mean ROUGE-L = 0.31 vs. 0.845 for templates). SAD achieves +12.0pp on human queries (CatR@1: 18.8% → 30.8%), a 63.8% relative gain—comparable to the 60% relative gain on templates. This cross-setting consistency provides strong evidence that SAD’s decomposition benefit generalizes beyond the template-specific text overlap.

H DAG Planner Pilot Evaluation

The DAG planner is an *optional extension* to the core decompose-retrieve pipeline; the main CatR@1 results in §7 do not depend on it. We include this pilot evaluation to demonstrate the feasibility of automatic dependency detection, while acknowledging that it requires further development before being considered a core contribution.

Relationship to main metrics. The DAG planner operates *downstream* of skill selection: it takes the already-selected skill sequence and adds dependency edges between them. Therefore, it does not affect CatR@1 or Chain_{cat} (which measure whether the *correct skills* are selected). Its contribution is orthogonal: improving *execution ordering* rather than *skill selection accuracy*. On the 20-query pilot, the DAG planner identifies at least one dependency edge in 80% of queries, and the overall Coherence score (0.54) indicates that roughly half of detected dependencies are functionally valid—suggesting room for improvement but demonstrating the feasibility of automatic dependency detection.

Table 9 reports the DAG planner evaluation on a stratified 20-query pilot (8 easy, 8 medium, 4 hard) using Qwen2.5-7B + SAD ($H=15$).

Difficulty	n	Step Acc.	Edges/q	Coherence	Latency (ms)
Easy	8	0.75	0.8	0.56	849
Medium	8	0.75	1.6	0.56	564
Hard	4	0.50	2.5	0.44	889
Overall	20	0.70	1.4	0.54	743

Table 9: DAG planner pilot evaluation. Step Acc.: fraction of queries where predicted step count matches ground truth. Edges/q: average dependency edges detected per query. Coherence: functional coherence (fraction of adjacent skill pairs with compatible I/O categories). 80% of queries have at least one detected dependency edge. Median latency is 568ms (p95: 1,261ms). Latency variation across difficulty levels reflects small sample sizes ($n=4-8$) rather than a systematic trend.

I SAD Prompt Templates

System prompt (shared by vanilla and SAD).

You are a task decomposition assistant. Given a complex user query, break it down into atomic sub-tasks, each requiring exactly one tool or skill. Output a JSON array of strings. Each string should be a concise, actionable sub-task description.

Vanilla user prompt.

```
Decompose the following query  
into atomic sub-tasks:  
{query}
```

SAD user prompt (Pass 2).

```
Decompose the following query  
into atomic sub-tasks.  
Available skills that may be  
relevant: {hint_list}  
Query: {query}
```

where $\{hint_list\}$ is a comma-separated list of the top- H skill names retrieved in Pass 1 (see Algorithm 1).